

**1. Parallelogram law of forces.** It states as under :

*“If two forces, acting simultaneously on a particle, be represented in magnitude and direction by the two adjacent sides of a parallelogram then their resultant may be represented in magnitude and direction by the diagonal of the parallelogram which passes through their point of intersection.”*

Refer Fig. 2.14. Let two forces  $P$  and  $Q$  acting simultaneously on a particle be represented in magnitude and direction by the adjacent sides  $oa$  and  $ob$  of a parallelogram  $oacb$  drawn from a point  $o$ , their resultant  $R$  will be represented in magnitude and direction by the diagonal  $oc$  of the parallelogram.

The value of  $R$  can be determined either graphically or analytically as explained below :

**Graphical method.** Draw vectors  $oa$  and  $ob$  to represent to some convenient scale the forces  $P$  and  $Q$  in magnitude and direction. Complete the parallelogram  $oacb$  by drawing  $ac$  parallel to  $ob$  and  $bc$  parallel to  $oa$ . The vector  $oc$  measured to the same scale will represent the resultant force  $R$ .

**Analytical method.** As shown in Fig. 2.14, in the parallelogram  $oacb$ , from  $c$  drop a perpendicular  $cd$  to  $oa$  at  $d$  when produced. Now from the geometry of the figure.

$$\angle cad = \theta, ac = Q$$

$$\therefore cd = Q \sin \theta$$

$$\text{and } ad = Q \cos \theta$$

From right-angled triangle,  $odc$

$$oc = \sqrt{(od)^2 + (cd)^2}$$

$$= \sqrt{(oa + ad)^2 + (cd)^2}$$

$$\text{or } R = \sqrt{(P + Q \cos \theta)^2 + (Q \sin \theta)^2}$$

$$= \sqrt{P^2 + Q^2 \cos^2 \theta + 2PQ \cos \theta + Q^2 \sin^2 \theta}$$

$$= \sqrt{P^2 + Q^2 (\sin^2 \theta + \cos^2 \theta) + 2PQ \cos \theta}$$

$$= \sqrt{P^2 + Q^2 + 2PQ \cos \theta} \quad (\because \sin^2 \theta + \cos^2 \theta = 1)$$

$$\therefore R = \sqrt{P^2 + Q^2 + 2PQ \cos \theta} \quad \dots(2.3)$$

Let the resultant makes an angle  $\alpha$  with  $P$  as shown in figure.

$$\begin{aligned} \text{Then } \tan \alpha &= \frac{cd}{od} = \frac{cd}{oa + ad} \\ &= \frac{Q \sin \theta}{P + Q \cos \theta} \end{aligned} \quad \dots(2.4)$$

**Case 1.** If  $\theta = 0^\circ$ , i.e., when the forces  $P$  and  $Q$  act along the same straight line then equation (2.3) reduces to

$$R = P + Q \quad (\because \cos 0^\circ = 1)$$

**Case 2.** If  $\theta = 90^\circ$ , i.e., when the forces  $P$  and  $Q$  act at right angles to each other, then

$$R = \sqrt{P^2 + Q^2} \quad (\because \cos 90^\circ = 0)$$

**Case 3.** If  $\theta = 180^\circ$ , i.e., the forces  $P$  and  $Q$  act along the same straight line but in opposite directions, then

$$R = P - Q \quad (\because \cos 180^\circ = -1)$$

The resultant will act in the direction of the greater force.

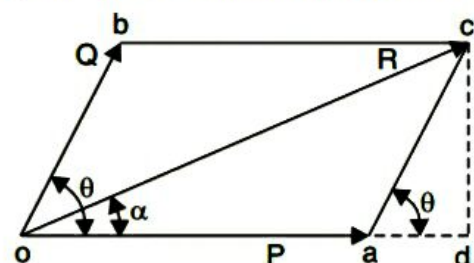


Fig. 2.14

## 2.16. LAMI'S THEOREM

It states as under :

*“If three coplanar forces acting on a point in a body keep it in equilibrium, then each force is proportional to the sine of the angle between the other two forces.”*

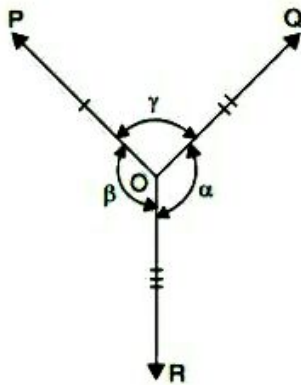


Fig. 2.28

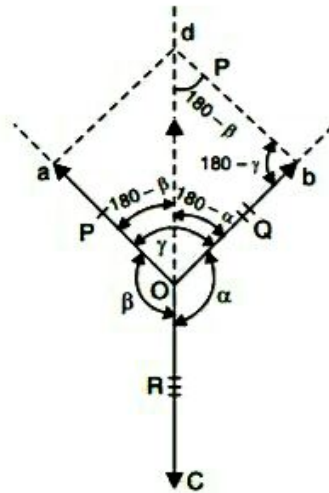


Fig. 2.29

Fig. 2.28 shows three forces  $P$ ,  $Q$  and  $R$  acting at a point  $O$ . Let the angle between  $P$  and  $Q$  be  $\gamma$ , between  $Q$  and  $R$  be  $\alpha$  and between  $R$  and  $P$  be  $\beta$ . If these forces are in equilibrium then according to Lami's theorem :

$$\frac{P}{\sin \alpha} = \frac{Q}{\sin \beta} = \frac{R}{\sin \gamma} \quad \dots(2.8)$$

**Proof.** Let us first consider the two forces  $P$  and  $Q$  which are represented by the two sides  $oa$  and  $ob$  of a parallelogram  $oadb$  as shown in Fig. 2.29. Then the resultant of these two forces will be given by  $od$  (the diagonal of the parallelogram) in magnitude and direction. This means  $od$  should be equal to  $R$  in magnitude but opposite in direction to  $oc$  as  $P$ ,  $Q$  and  $R$  are in equilibrium.

From geometry of parallelogram  $oadb$ , we find

$$bd = P \text{ and } ad = Q$$

$$\angle bod = (180 - \alpha)$$

and

$$\angle bdo = \angle aod = \angle (180 - \beta)$$

Now, from triangle  $obd$ ,

$$\angle obd = 180^\circ - \angle bod - \angle bdo$$

$$(\because \text{Sum of all the angles of triangle} = 180^\circ)$$

$$= 180 - (180 - \alpha) - (180 - \beta) = \alpha + \beta - 180^\circ$$

But

$$\alpha + \beta + \gamma = 360^\circ$$

$$(\because \text{Sum of all the angles at a point} = 360^\circ)$$

Subtracting  $180^\circ$  from both the sides, we get

$$\alpha + \beta + \gamma - 180^\circ = 360^\circ - 180^\circ$$

$$(\alpha + \beta - 180^\circ) + \gamma = 180^\circ$$

$$\angle obd + \gamma = 180^\circ$$

$$\angle obd = 180 - \gamma$$

Now, applying sine formula to triangle  $obd$ ,

$$\frac{bd}{\sin(180 - \alpha)} = \frac{ob}{\sin(180 - \beta)} = \frac{od}{\sin(180 - \gamma)}$$

$$\frac{P}{\sin(180 - \alpha)} = \frac{Q}{\sin(180 - \beta)} = \frac{R}{\sin(180 - \gamma)}$$

Hence

$$\frac{P}{\sin \alpha} = \frac{Q}{\sin \beta} = \frac{R}{\sin \gamma}.$$

### 3.5. VARIGNON'S THEOREM

It states as under :

*"The algebraic sum of the moments of two forces about any point in their plane is equal to the moment of their resultant about that point."*

**Proof.**

**Case 1.** When the two forces meet at a point.

Fig. 3.6 shows two forces  $P$  and  $Q$  acting at  $A$ . The magnitude of  $P$  is represented by  $AB$  and that of  $Q$  is represented by  $AD$ . Complete the parallelogram  $ABCD$ .  $AC$  represents the resultant  $R$  of  $P$  and  $Q$ .

Take any other point  $O$  in the plane of the forces  $P$  and  $Q$  and in the line  $CD$  produced as shown. Join  $OB$  and  $OA$ .

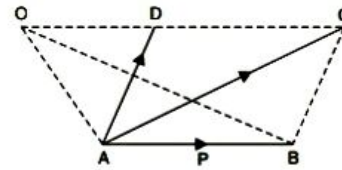


Fig. 3.6

$$\text{Moment of } P \text{ about } O = 2 \Delta OAB$$

$$\text{Moment of } Q \text{ about } O = 2 \Delta OAD$$

$$\text{Moment of } R \text{ about } O = 2 \Delta OAC$$

$$\text{But area of } \Delta OAB = \text{area of } \Delta ABC = \text{area of } \Delta ACD$$

$$\text{Adding algebraically the moments of } P \text{ and } Q$$

$$= 2 \Delta OAB + 2 \Delta OAD$$

$$= 2 \Delta ACD + 2 \Delta OAD$$

$$(\text{Substituting } 2\Delta ACD \text{ for } \Delta OAB \text{ which are equal})$$

$$= 2 (\Delta ACD + \Delta OAD)$$

$$= 2 \Delta OAC$$

$$= \text{Moment of } R \text{ about } O.$$

**Note.** Refer Fig. 3.7. Consider a force  $P$  which can be represented in magnitude and direction by the line  $AB$ . Let  $O$  be the point, about which the moment of this force is required. Draw  $OM$  perpendicular to  $AB$  and join  $OA$  and  $OB$ .

$$\text{Now moment of force } P \text{ about } O = P \times OM$$

$$= AB \times OM$$

But  $AB \times OM$  is equal to twice the area of triangle  $OAB$  because geometrically the area of this triangle is equal to  $(AB \times OM)/2$ .

$$\therefore \text{Moment of force } P \text{ about } O \text{ i.e., } AB \times OM = 2 \Delta AOB.$$

**Case 2.** When the two forces are parallel to each other :

Let  $P$  and  $Q$  be the two parallel forces as shown in Fig. 3.8. Draw a line  $AB$  perpendicular to the forces to meet their lines of action in  $A$  and  $B$ . Locate any point  $O$  in the plane of the two forces on  $AB$  produced. The resultant of  $P$  and  $Q$  will be  $R$  which is equal to sum of forces  $P$  and  $Q$ . Let it act through a point  $C$  in  $AB$  so that

$$Q \times CB = P \times CA$$

The sum of moments of  $P$  and  $Q$  about  $O$

$$= P \times OA + Q \times OB$$

$$= P(OC + CA) + Q(OC - CB)$$

$$= (P + Q) OC + P \times CA - Q \times CB$$

$$= (P + Q) OC$$

$$= \text{Moment of } R \text{ about } O.$$

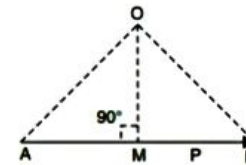


Fig. 3.7

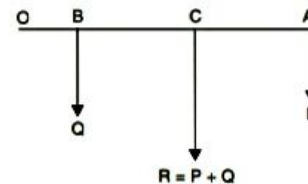


Fig. 3.8

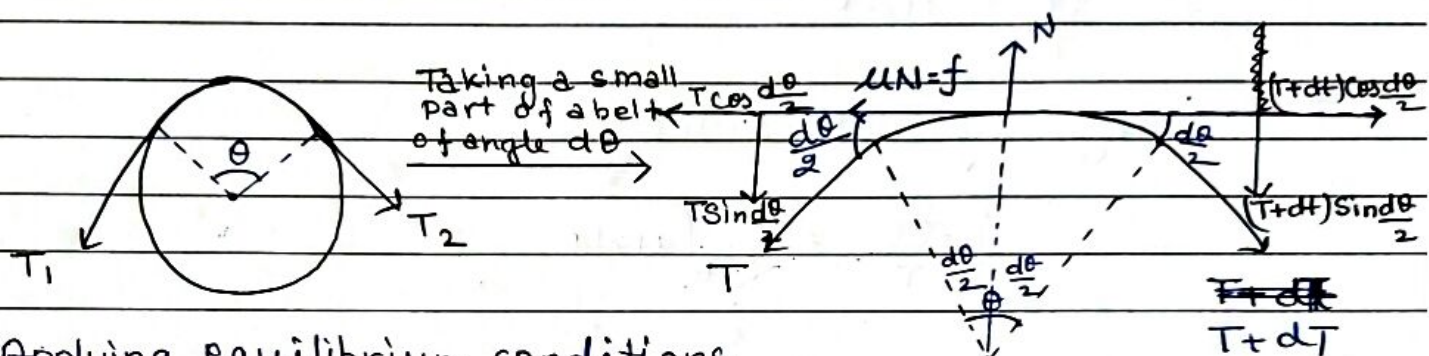
$$(\because Q \times CB = P \times CA)$$

**Note.** Varignon's theorem is applicable to cases, where the two forces give out a single resultant and it does not apply when the forces form a couple since the resultant force of a couple is zero.

The principle of moments (discussed earlier) is the extension of Varignon's theorem.

## Belt Friction

→ Derivation of Ratio of belt tensions on the tight side and slack side for a flat belt



Applying equilibrium conditions

$$\sum F_x = 0$$

$$\Rightarrow (T+dT)\cos\frac{d\theta}{2} - T\cos\frac{d\theta}{2} - \mu N = 0$$

$$\Rightarrow T\cos\frac{d\theta}{2} + \frac{dT\cos\frac{d\theta}{2}}{2} - T\cos\frac{d\theta}{2} = \mu N$$

$$\Rightarrow dT\cos\frac{d\theta}{2} = \mu N$$

$$\Rightarrow dT = \mu N \quad \text{--- (i)}$$

$$\Rightarrow N = \frac{dT}{\mu}$$

$$\left[ \begin{array}{l} \because \theta \text{ is very small} \\ \therefore \cos\frac{d\theta}{2} \approx \cos 0 = 1 \end{array} \right]$$

$$\sum F_y = 0$$

$$\Rightarrow N - (T+dT)\sin\frac{d\theta}{2} - T\sin\frac{d\theta}{2} = 0$$

$$\Rightarrow N - T\sin\frac{d\theta}{2} - \frac{dT\sin\frac{d\theta}{2}}{2} - T\sin\frac{d\theta}{2} = 0$$

$$\Rightarrow N - 2T\sin\frac{d\theta}{2} - \frac{dT\sin\frac{d\theta}{2}}{2} = 0$$

$$\Rightarrow N - \cancel{2T}\frac{d\theta}{2} - \frac{dT d\theta}{2} = 0$$

$$\left[ \begin{array}{l} \because \theta \text{ is very small} \\ \therefore \sin\frac{d\theta}{2} \approx \frac{d\theta}{2} \end{array} \right]$$

$$N = T d\theta \quad \text{--- (ii)}$$

$\left[ \because dT \text{ \& } d\theta \text{ both are very small \& small.} \right]$

$$\frac{dT}{\mu} = T d\theta \quad (\text{From eqn (i)})$$

$$\therefore dT \cdot \frac{d\theta}{\mu} = 0$$

$$\Rightarrow \frac{dT}{T} = \mu d\theta$$

Integrating both sides,

$$\int_{T_2}^{T_1} \frac{dT}{T} = \int \mu d\theta$$

$$\Rightarrow \left[ \ln T \right]_{T_2}^{T_1} = \mu \theta$$

$$\Rightarrow \ln T_1 - \ln T_2 = \mu \theta$$

$$\Rightarrow \ln \left( \frac{T_1}{T_2} \right) = \mu \theta$$

$$\Rightarrow \frac{T_1}{T_2} = e^{\mu \theta}$$

Hence,

Ratio of belt tensions on the tight side and slack side for a flat belt is,

$$\boxed{\frac{T_1}{T_2} = e^{\mu \theta}}$$

Where,

$T_1$  = Tension on

$T_2$  = Tension on

$\mu$  = coefficient of friction

$\theta$  = angle b/w contact surfaces.

## # Newton's Second Law of Motion

Newton's second law of motion states, "The rate of change of momentum is directly proportional to the impressed force, and takes place in the same direction in which the force acts".

Let,  $m$  = mass of a body,  
 $u$  = initial velocity,  
 $v$  = final velocity,  
 $a$  = constant acceleration,  
 $t$  = time required to change the velocity from  $u$  to  $v$ ; and  
 $P$  = force required to change velocity from  $u$  to  $v$  in  $t$  time.

Initial momentum of the body =  $mu$

Final momentum of the body =  $mv$

Change of momentum of the body =  $mv - mu$

$$\begin{aligned}\text{Rate of change of momentum of the body} &= \frac{mv - mu}{t} \\ &= \frac{m(v - u)}{t} \\ &= ma \quad \left\{ \because \frac{v - u}{t} = a \right\}\end{aligned}$$

According to Newton's second law of motion,

$$P \propto ma$$

$$P = kma$$

where  $k$  is a constant of proportionality  
if we define unit force as that which produce unit acceleration in unit mass, we get by putting

$$P = 1, m = 1, a = 1$$

$$k = 1$$

Hence,  $\boxed{P = ma}$

### Work Energy Principle :-

Work done by force acting on a body during its displacement is equal to change in its kinetic energy.

$$\text{Work Done} = \text{Change in K.E.}$$

$$F \times S = \frac{1}{2} m (v^2 - u^2)$$

$$F \times S = \frac{1}{2} \times \frac{W}{g} \times (v^2 - u^2)$$

By using Newton's 2<sup>nd</sup> law

$$F = ma$$

$$F = m \times \frac{dv}{dt}$$

$$F = m \times \frac{dv}{dt} \times \frac{ds}{ds}$$

$$F = m \times \frac{dv}{ds} \times \frac{ds}{dt}$$

$$F \times ds = m \times dv \times \frac{ds}{dt}$$

$$F \times ds = m \times dv \times v$$

$$\int_{s_1}^{s_2} F \times ds = \int_u^v m v dv$$

$$F [s]_{s_1}^{s_2} = m \left[ \frac{v^2}{2} \right]_u^v$$

$$F [s_2 - s_1] = \frac{1}{2} m [v^2 - u^2]$$

$$F \times S = \frac{1}{2} m (v^2 - u^2)$$

where ,  
F = force  
S = displacement  
v = final velocity  
u = initial velocity

## Impulse - Momentum Principle

→ "Change in momentum of body or system is equal to the impulse or impulsive force."

Initial momentum  $\Rightarrow mv_1$

final momentum  $\Rightarrow mv_2$

$\therefore$  change in momentum  $\Rightarrow$  final - Initial  
momentum momentum

$$\Rightarrow mv_2 - mv_1$$

As per the Impulse Momentum Principle -

Change in Momentum  $\Rightarrow$  Impulse

$$mv_2 - mv_1 = \int_{t_1}^{t_2} F dt$$

### 8.9. D' ALEMBERT'S PRINCIPLE

D' Alembert, a French mathematician, was the first to point out that on the lines of *equation of static equilibrium, equation of dynamic equilibrium can also be established by introducing inertia force in the direction opposite the acceleration in addition to the real forces on the plane.*

Static equilibrium equations are :

$$\Sigma H \text{ (or } P_x) = 0, \Sigma V \text{ (or } \Sigma P_y) = 0, \Sigma M = 0$$

Similarly when different external forces act on a system in motion, the algebraic sum of all the forces (including the *inertia force*) is zero. This is explained as under :

We know that,  $P = ma$  (Newton's second law of motion)

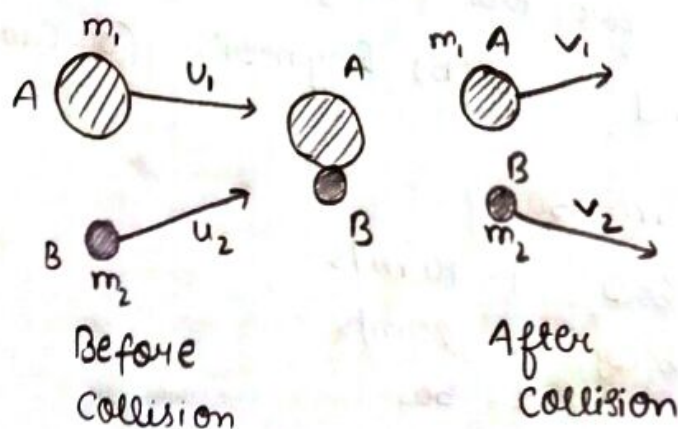
or  $P - ma = 0$  or  $P + (-ma) = 0$

The expression in the block  $(-ma)$  is the *inertia force* and negative sign signifies that it acts in a direction opposite to that of acceleration/retardation  $a$ .

It is also known as the "*principle of kinostatics*".

## # Law of Conservation of Momentum

The law of conservation of momentum states: "Total momentum of any group of objects always remains the same if no external force acts on them."



Total momentum before collision:

Momentum of mass,  $m_1 = m_1 u_1$

Momentum of mass,  $m_2 = m_2 u_2$

Total momentum of masses  $m_1$  and  $m_2 = m_1 u_1 + m_2 u_2$

Total momentum after collision,

Momentum of mass,  $m_1 = m_1 v_1$

Momentum of mass,  $m_2 = m_2 v_2$

$\therefore$  Total momentum of masses  $m_1$  and  $m_2 = m_1 v_1 + m_2 v_2$

According to law of conservation of momentum

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

e.g :- (i) Propulsion of a jet plane.

(ii) Recoil of a gun.

# Principle of Conservation of Energy

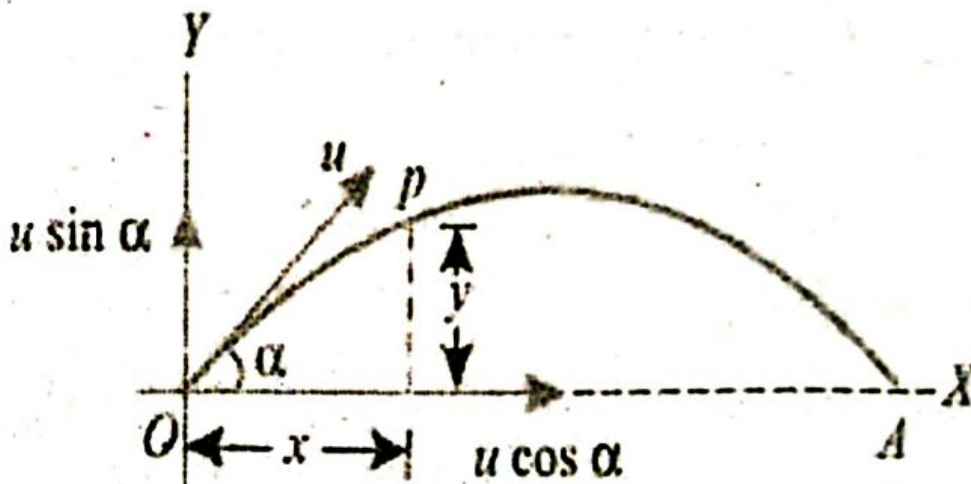
$$\boxed{\Delta K \cdot E \Rightarrow \Delta P \cdot E}$$

$$(K \cdot E)_2 - (K \cdot E)_1 \Rightarrow (P \cdot E)_1 - (P \cdot E)_2$$

$$\boxed{(K \cdot E)_1 + (P \cdot E)_1 \Rightarrow (K \cdot E)_2 + (P \cdot E)_2}$$

**Q. Derive the equation of a path of a projectile.**

**Ans.**



Consider a particle projected from a point  $O$  at a certain angle with the horizontal. As already discussed, the particle will move along certain path  $OPA$ , in the air, and will fall down at  $A$  as shown in Fig.

Let  $u$  = Velocity of projection, and

$\alpha$  = Angle of projection with the horizontal.

Consider any point  $P$  as the position of particle, after  $t$  seconds with  $x$  and  $y$  as co-ordinates as shown in Fig. We know that horizontal component of the velocity of projection.

$$= u \cos \alpha$$

$$= u \sin \alpha$$

and vertical component

$$\therefore y = u \sin \alpha t - \frac{1}{2} g t^2$$

$$x = u \cos \alpha t$$

$$\text{or, } t = \frac{x}{u \cos \alpha}$$

Substituting the value of  $t$  in equation (i),

$$y = u \sin \alpha \left( \frac{x}{u \cos \alpha} \right) - \frac{1}{2} g t^2$$

and

$$x = u \cos \alpha t$$

$$t = \frac{x}{u \cos \alpha}$$

Substituting the value of  $t$  in equation (i),

$$\begin{aligned} y &= u \sin \alpha \left( \frac{x}{u \cos \alpha} \right) - \frac{1}{2} g \left( \frac{x}{u \cos \alpha} \right)^2 \\ &= x \tan \alpha - \frac{g x^2}{2 u^2 \cos^2 \alpha} \end{aligned}$$

Since this is the equation of a parabola, therefore path of a projectile (or the equation of trajectory) is also a parabola.

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